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Environmental policies and productivity growth: Evidence across industries and firms

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ABSTRACT

This paper investigates the impact of changes in environmental policy stringency on industry- and firm-level productivity growth in a panel of OECD countries. To test the strong version of the Porter Hypothesis (PH), we extend a neo-Schumpeterian productivity model to allow for effects of environmental policies. We use a new environmental policy stringency (EPS) index and let the effect of countries' environmental policies vary with the pollution intensity of the industry and with the countries' and firms' technological advancement. A tightening of environmental policy is associated with a short-term increase in industry-level productivity growth in the most technologically-advanced countries. This effect diminishes with the distance to the global productivity frontier, eventually becoming insignificant. For the average firm, no evidence of PH is found. However, the most productive firms see a temporary boost in productivity growth, while the less productive ones experience a productivity slowdown.

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Introduction

Over the past twenty years, governments in OECD countries have implemented a wide range of environmental policies with the aim of improving environmental conditions. Policy instruments are often based on price mechanisms (market-based instruments), increasing the opportunity costs of pollution and environmental damage, or take the form of command-and-control policies, enforcing environmental standards (non-market instruments). Inevitably, environmental policies affect production processes, resource reallocation, capital investment, labor intensity and innovation incentives. Therefore, the ongoing tightening of environmental regulation is likely to affect not only environmental outcomes but also economic performance.

This paper focuses on the effects of environmental policy stringency on multi-factor productivity (MFP) growth. More stringent environmental policies imply an additional burden for firms, inducing a shift of resources from the traditionally "productive" uses towards pollution abatement. Hence, firm-level productivity growth may slow down, at least in the short

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term. At the same time, incentives for innovation, efficiency improvements and within-firm reallocation may lead to higher productivity, as suggested by the so-called Porter Hypothesis (PH; Porter, 1991; Porter and van der Linde, 1995). Jaffe and Palmer (1997) characterize three variants of the Porter Hypothesis: the *weak* version implies that tighter environmental policies stimulate innovation, whereas the *strong* version claims that environmental policy may lead to higher overall productivity of firms. Finally the *narrow version* of the PH states that gains in productivity and innovation redirection are more likely under certain types of environmental policies, such as flexible, market-based instruments. At the industry level, the direction of the effect of environmental policies on productivity will also depend on the resulting resource reallocation among firms: polluting firms may exit or enter the market, decrease or increase their production or outsource part of their activities or relocate abroad (in line with the pollution haven hypothesis, see for example Levinson and Taylor (2008)). Furthermore, if environmental policies increase barriers to entry, competition or trade, competitive pressures will be reduced, firms' entry-exit dynamics will slow and industry productivity will decrease. Hence, understanding the impact of environmental policies on MFP growth is crucial for the design and choice of policy packages. Yet, empirical evidence on such effects has been rather country- and context-specific and thus inconclusive. The two main constraints on empirical work so far are poor data availability, as good cross-country proxies for environmental policy stringency are hard to come by (Botta and Kozluk, 2014; Brunel and Levinson, 2016), and the fact that identification strategies based on the introduction of one policy in several countries at the same time are not feasible due to a scarcity of such examples and multiple interfering factors.

In an attempt to fill this gap, we focus on the strong version of the PH and conduct an empirical analysis of the effect of environmental policy stringency (EPS) on productivity growth. The contribution of this analysis is twofold. First, we employ a new cross-country proxy of environmental policy stringency, developed by Botta and Kozluk (2014).[3] This allows a panel analysis of OECD countries over two decades giving a more global perspective to the literature on the Porter Hypothesis. Second, we combine industry and firm level results. The industry analysis considers aggregate effects on the full firm population, that is, including entry and exit and the relative importance of effects across firms weighted by their size, but may suffer from aggregation bias: different effects of environmental policy stringency on firm-level MFP growth may cancel out on the industry level. The firm level analysis can address this heterogeneity directly, but has limitations in terms of population representativeness and tracking firm entry and exit dynamics. Thus a combination of analysis at the two levels offers additional insights on the channels at work.

The results suggest that a tightening in environmental policies has a positive short-term effect on industry productivity growth in the most technologically advanced country-industry pairs. This positive effect diminishes with the distance to the global technology frontier and vanishes completely for the least productive ones. This finding is only partially reflected in the results at the firm level: only one-fifth of the firms are able to reap productivity gains after a tightening of environmental regulation. About half of the firms, the least productive ones, face a negative effect on productivity growth in the short run. This negative effect, for less technologically advanced firms, is lost at the industry level due to aggregation. Moreover, empirical results across the two levels of analysis yield some support for the narrow version of the Porter Hypothesis: market-based environmental policies are found to be more productivity-friendly than non-market instruments.

The paper is structured in the following way. Section 2 provides an overview of the relevant empirical literature on the effects of environmental policies on productivity growth. Section 3 presents the EPS index used in the analysis, followed by Section 4, providing some descriptive statistics. Section 5 explains the empirical methodology, details about the data used, the main results, and discusses potential channels of the effects. Section 6 provides additional tests and Section 7 robustness checks. The final section concludes.

Literature review

Empirical studies about the PH can be broadly divided into two main strands: analyses of the "strong" version look at productivity growth, while analyses of the "weak" version focus on innovation. The "narrow" version is less frequently analysed.

At the industry level, empirical studies of the effects of environmental policies on productivity growth provide rather inconclusive evidence on the significance and direction of the effect (Cohen and Tubb, 2015; Kozluk and Zipperer, 2014; Ambec et al., 2013). This may be because they often analyze rather context specific set-ups, such as single countries, industries or specific environmental laws. Older industry-level studies, which tend to find negative effects of environmental regulations on productivity growth, suffer from problems of identification and are generally not very robust. Notable contributions include attempts to explain the US productivity slowdown in the 1970s with environmental regulation (Gray, 1987; Barbera and McConnell, 1990), as well as Dufour et al. (1998) who looks at Canadian industries in the 1980s. More recent work, based on longer time series or case studies, finds an aggregate positive or nil effect. Hamamoto (2006) finds that environmental command and control regulations led to an increase in innovation (R&D spending) and consequently to an increase in productivity growth in a sample of five Japanese manufacturing sectors over 20 years. Yang et al. (2012) find similar effects of environmental policy tightening on Taiwanese manufacturing sectors; while for Quebec, Lanoie et al. (2008) find a short-run negative effect, due to the additional costs imposed by the tighter environmental regulation, outweighed by a subsequent positive effect. An early study with an

[3] The OECD's Environmental Policy Stringency indicator was developed as part of the same project as this empirical work. Details on the indicator, updates and the underlying data can be found at http://oe.cd/eps.

international dimension, Alpay et al. (2002), finds no effect on productivity growth in the US food manufacturing sector (where environmental policies are proxied by pollution abatement and control expenditures) but a positive effect in Mexico (with environmental policies proxied by the frequency of environmental inspections). More recently, Franco and Marin (2015) find a positive effect of environmental taxes on productivity in European countries in the early 2000s, but no effect on innovation (measured by patent counts), suggesting a direct effect of environmental policies. Rubashkina et al. (2015) find the opposite effects in a panel of 17 European manufacturing sectors using abatement cost data as the environmental policy proxy: industry productivity seems unaffected by abatement costs whereas patenting activity increased.

Evidence from firm or plant level studies shows a negative but also not very robust effect of environmental regulation on productivity growth (Cohen and Tubb, 2015; Kozluk and Zipperer, 2014). Most of the studies compare productivity growth between regulated and non-regulated firms or plants, finding negative (Gollop and Roberts, 1983; Smith and Sims, 1985) or insignificant (Berman and Bui, 2001) results. These early studies often neglect including firm or plant specific characteristics in the analysis. Becker (2011) and Gray and Shadbegian (2003) show the importance of including such characteristics and find negative effects of environmental regulation on productivity growth. The effect of environmental regulation can also depend on the type of pollutant regulated. In this respect, Greenstone et al. (2012) identify a negative effect of ozone and particulates emission regulations, no effect of sulfur dioxide emission regulations and a positive effect of carbon monoxide regulations. Aside the aforementioned methodological issues, the firm and plant level studies suffer from a lack of generality, as they usually analyze very specific regulations or industries in a single country setting.

Empirical studies on the Porter Hypothesis, strong or weak version, rely on a vast range of measures of environmental policy stringency. Most studies focusing on the effects in a single country look at the introduction, or significant change, in a particular environmental policy, or the difference between regulated and unregulated firms. Such approaches can be more precise in capturing causal relationships, but at the sacrifice of the generality of conclusions. Cross country (or state) studies are limited by the availability of cross-country measures of environmental policies. The proxies used range from pollution abatement expenditures (Gray, 1987; Morgenstern et al., 2002; Gray and Shadbegian, 2003), survey-based policy perceptions (Kalamova and Johnstone, 2011; Johnstone et al., 2010; Johnstone and Labonne, 2006; Lanoie et al., 2011), environmental treaties signed (Smarzynska and Wei, 2004; Yörük and Zaim, 2005; Wu and Wang, 2008), "green" voting records (Gray, 1997) or environment-related inspection frequency (Alpay et al., 2002; Testa et al., 2011; Brunnermeier and Cohen, 2003). In addition, some papers use revealed performance measures, such as energy or pollution intensity or environmental compliance (Cole and Elliot, 2003; Van Beers and Van den Bergh, 1997; Harris et al., 2003; McConnell and Schwab, 1990; Smarzynska and Wei, 2004).

EPS – a new cross-country measure of environmental policy

Our paper's contribution is to provide a cross-country panel analysis of the strong version of Porter's Hypothesis using a new composite index of environmental policy stringency (EPS) developed by the OECD (Botta and Kozluk, 2014). The EPS index covers 24 OECD countries over the period 1990–2012 and summarizes environmental policy stringency across selected instruments. The indicator is based on the taxonomy developed by De Serres et al. (2010) and consists of two components, a market-based and a non-market one. The market-based component groups instruments which assign an explicit price to the externalities (taxes: CO_2, SO_X, NO_X, and diesel fuel; trading schemes: CO_2, renewable energy certificates, energy efficiency certificates, feed-in-tariffs, and deposit-refund-schemes), while the non-market component clusters command-and-control instruments, such as standards (emission limit values for NO_X, SO_X, and PM, limits on sulfur content in diesel), and technology-support policies, such as government R&D subsidies. The indicator ranges from 0 to 6 with higher numbers being associated with more stringent environmental policies. The novelty and advantage of this indicator is that it reduces a complex set of multi-dimensional policies into a comparable country-specific proxy. While the indicator focusses on regulation in upstream activities, such as energy and transport, and covers primarily climate and air pollution policies, the intention is to proxy overall environmental policy stringency. Botta and Kozluk (2014) argue that the advantage of focusing on policies affecting upstream activities is that such activities are present and of comparable economic importance across countries. Both energy and transport are generally characterized by high pollution intensity and regulated by well-identified and comparable types of policy instruments. The key idea is that the stringency in this subsample of sectors and pollutants implies a similar degree of policy control for the same externalities in other sectors.[4] Moreover, from an econometric perspective, using an EPS proxy based on sectors that are not directly included in the analysis (such as energy, transport and waste) helps to avoid some endogeneity concerns.[5] In practice, the credibility of the EPS as proxy for the country's overall environmental policy stringency is confirmed by a comparison with other available measures of environmental policy stringency (e.g. World Economic Forums Executive Opinion Survey responses, EBRD's CLIMI).[6]

[4] An idea that seems confirmed by Rubashkina et al. (2015).

[5] See section "Robustness checks" for a detailed discussion.

[6] Three main cross-country measures of environmental policies are available: responses to World Economic Forum's Executive Opinion Survey (on perceived stringency), a composite indicator based on different policy aspects (Dasgupta et al., 1995) and the EBRD's Climate Laws, Institutions and Measures Index (CLIMI, EBRD, 2011). These indicators either lack a meaningful time-dimension, or lack one at all, which renders them less suitable for empirical analyses of effects of policy stringency (see Botta and Kozluk (2014) and Brunel and Levinson (2016) for a detailed discussion). Correlations for the EPS reported by Botta and Kozluk (2014) are: with WEF 0.44 (p-value 0.00), with CLIMI 0.56 (p-value 0.01).

Environmental policies and productivity – what the descriptive statistics say

Over the period of 1990–2009, countries with the most stringent environmental policies have generally seen higher industry MFP growth than those with the most lax policies. Fig. 1, left panel, shows that the median of annual industry-level MFP growth rates for the most stringent countries is on average higher (i.e. located more to the right) than for the less stringent countries. A similar finding holds when looking at tightening in environmental policies (Fig. 1, right panel). Countries that experienced a stronger tightening have, on average, experienced higher MFP growth than countries with less tightening. While this would suggest that environmental policy stringency is associated with higher productivity growth, it can only serve as a motivation for a rigorous analysis, as many factors can influence productivity. Data section and Appendix A provide additional information and statistics of the main variables of interest.

Econometric model

Our empirical analysis is based on a standard Neo-Schumpeterian model of multifactor productivity growth, where productivity growth depends on the industry's (or firm's) ability to adopt innovative and efficient technologies available in the market (technological catch-up) and on firms' ability to innovate (technological pass-through) (Acemoglu et al., 2006; Aghion and Howitt, 2006; Nicoletti and Scarpetta, 2003). As stressed in a more general context by Bourlès et al. (2013), regulations may have a heterogeneous effect on firm and industry productivity growth depending on the level of countries' and firms' technological advancement. A similar distinction is suggested by Porter and van der Linde (1995) for effects of environmental policies. Following this argument, we augment the model of Bourlès et al. (2013) with environmental policies, allowing the latter to have different effects on MFP growth depending on the country's/firm's distance to the global frontier. The most technologically advanced firms are likely to have more resources to invest into R&D or knowledge-based capital, and scale up energy efficiency gains from abatement. They may also be better suited to adapt to changes in the business environment – in this case provoked by an EPS tightening – having better access to technology, markets or managerial capacity. Less advanced firms may find it burdensome to comply with the new regulation, and may require higher investments to adopt cleaner technologies, exchanging equipment, etc.

The identification strategy is based on the fact that environmental policy is likely to affect industry productivity heterogeneously depending on the industry's exposure to the regulation. A higher 'environmental dependence' (proxied by pollution intensity in our main specification) increases the industries' exposure to the country-level environmental policies and hence the potential economic effects of the latter. Thus, in the empirical specification, the EPS index is interacted with the pollution intensity of the industry. This approach is common in the analysis of impacts of national-level policies and developments at the industry and firm levels, and was popularized by Rajan and Zingales (1998), who looked at the role of financial development using financial dependence as the exposure variable. See Data section and Appendix A for more details on the environmental dependence variable.

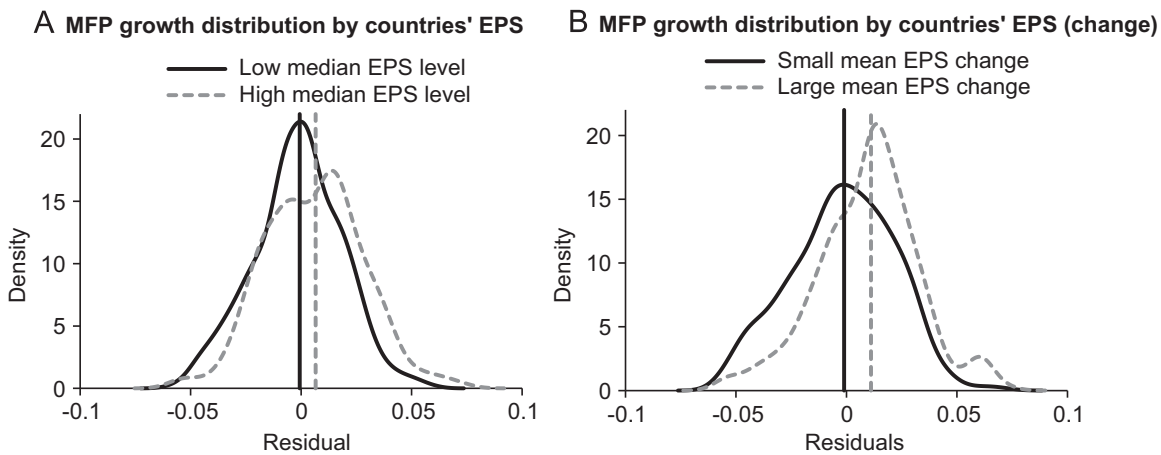

Fig. 1. Distribution of MFP growth by countries' environmental policy stringency (EPS) 1990–2009. *Note*: The left-hand graph presents the MFP growth distribution of the three countries with highest median EPS levels (dotted line) versus the distribution in the three countries with the lowest median EPS levels. The right-hand graph shows the MFP growth distribution of the three countries with the highest average EPS increases over the sample versus the three countries with the lowest.
Source: Authors' calculations based on OECD STAN database and OECD PDBi database.

The resulting specification for the industry and firm analysis is the following:

$$\Delta \ln MFP_{cit} = \alpha_1 + \alpha_2 \frac{1}{n} \sum_{j=1}^{n} \left(ED_{i1987} \Delta EPS_{ct-j}\right) + \alpha_3\, gap_{cit-1} \frac{1}{n} \sum_{j=1}^{n} \left(ED_{i1987} \Delta EPS_{ct-j}\right)$$
$$+ \alpha_4\, gap_{cit-1} + \alpha_5\, \Delta \ln \widehat{MFP}_{it} + \boldsymbol{x_{cit}\gamma} + \eta_t + \delta_{ci} + \epsilon_{cit}$$

where $\Delta \ln MFP_{cit}$ is the multifactor productivity growth for each combination of country c and industry/firm i at time t. ΔEPS is the change of the country EPS and captures the tightening of country's environmental policy interacted with pre-sample industry pollution intensity, ED, and used here as n-year moving average. The second term allows for a heterogeneous effect of the policy depending on the technological gap, defined as the distance to the country-industry frontier $gap = \ln\left(\frac{\widehat{MFP}_i}{\widehat{MFP}_{ci}}\right)$. The rest of the terms follow directly from Bourlès et al. (2013). The third term is the distance to the productivity frontier which allows for technological catch-up effects. The global frontier is defined as the highest MFP across countries by industry and year - corrected for outliers, following Bas et al. (forthcoming). In the firm-level estimation, the global frontier is defined as the average MFP of the top 5% firms across countries, by industry and year. The fourth term is the growth in the leader MFP, $\Delta \ln\widehat{MFP}_{it}$, and represents the technological pass-through. x_{cit} is a vector of additional country and industry/firm controls.

We use three different specifications in terms of controls. The 'Baseline' specification includes the output gap (to control for the business cycle), an additional dummy for particular effects of the financial crisis and a common time trend.[7] In the 'Baseline + Regulation' specification, we follow Bourlès et al. (2013) and Bassanini and Duval (2009) by including domestic regulations that are likely to affect MFP growth: employment protection legislation (OECD's EPL) and product market regulation (OECD's PMR). Higher employment protection may hinder the reallocation of workers across firms, thereby implying lower productivity growth (Bourlès et al., 2013), but it may also encourage investment in workers' human capital, boosting MFP growth (Gal and Theising, 2015). Anticompetitive product market regulations can hamper productivity by reducing competitive pressures, impeding efficient resource allocation and thereby raising production costs (Bourlès et al., 2013).[8] Thirdly, in the 'Full specification' we additionally control for the fact that R&D-intensive industries are more likely to have higher productivity growth, (including lagged R&D expenditure over value added). Finally, we control for a time trend η_t and country-industry fixed effects δ_{ci} or, alternatively, country and industry fixed effects: δ_c and δ_i.[9] At the firm level, we also control for firm specific factors by including standard controls such as demand proxied by total assets turnover, and firms' size (lagged log number of employees).[10]

The analysis focuses on the change of the EPS indicator rather than levels. Even though the level of environmental policy stringency may have economic effects, for instance on trade patterns, effects on productivity growth are likely to be driven by changes in the environmental policy stringency. When only considering one country, a static level of environmental regulation per se (seen as a difference in relative prices of inputs) will not induce changes in the production processes of firms. However, firms are likely to react to new policy implementations by investing into abatement capital or reorganizing their production. Moreover, from an econometric perspective, EPS levels empirically resemble a non-stationary process, which may be problematic, while the first differences are stationary.[11]

Effects of environmental policy changes can be lagged in time. The timing can depend on the type of instrument – e.g. it can be delayed with taxes, while immediate with bans or standards as well as its characteristics: announcement, derogations, delayed implementation of legislative processes, or specific characteristics of the reaction (time to build, learning by doing, etc.). Firms may invest into new capital, decide to exit the market or outsource, and hence see economic effects both before or after the policy's exact implementation date. Therefore, after testing for different lag structures, a three-year moving average has been chosen for the change in EPS and has been adopted for both the industry and firm level. This moving average is defined as the un-weighted average of the first, second and third lag of the change in EPS. This moving average is also used in the interaction term with the distance to frontier.[12]

[7] We test three alternative specifications to account for the structural breakdown due to the latest financial crises: a dummy equal to 1 from 2007 to 2009; three dummies, one for each year of crisis, and a single dummy in 2008 to capture beginning of the Euro crisis. Results are robust throughout these specifications.

[8] We distinguish between high and low regulated countries using a dummy variable which takes the value 1 for PMR levels above the median and 0 below. The distribution of the 1998 PMR is used to construct a dummy for 1990–2000, while the indicator values of 2003 and 2008 are attributed to the time periods 2001–2005 and 2006–2010, respectively, given that the PMR is only calculated every 5 years.

[9] Country-time fixed effects are not included in the analysis, but we use the mentioned comprehensive range of time and country-level controls.

[10] Given the unbalanced firm sample, we substitute the PMR dummy with the OECD regulatory impact (RI) indicator in the firm-level specification. While the PMR is an economy-wide indicator of product market policy regimes, the RI quantifies the potential knock-on effects of anti-competitive regulation in upstream network sectors on 39 sectors manufacturing sectors which rely on the network sector outputs as intermediate inputs. This indicator has the advantage of having more variability than the PMR, which makes it more suited for firm analysis where the time spell is short.

[11] We also test the impact of EPS in levels (in a similar setup), which turns out not to be significant.

[12] An introduction or a change in stringency of environmental policy instruments is often known or expected in advance – before the actual implementation. Thus the effect of a tightening in EPS on MFP growth in period t might not be captured by looking only at the following periods. If policymakers credibly announce the upcoming change of an environmental policy, firms may start investing into the capital and labor to comply with the new policy before the actual implementation. Tests on forward looking behavior are reported in the OECD working paper Albrizio et al. (2014), which also includes a macro-level analysis where there is some evidence of such effects.

Data

Industry MFP is constructed using the OECD Structural Analysis database (STAN) and the Productivity Database By industry (PDBi). It is calculated as the residual from a log Cobb–Douglas production function for capital and labor, with the capital share set to 1/3. MFP growth is estimated as a two-year moving average for a panel of 17 OECD countries and 10 manufacturing sectors over the time period 1990–2009.[13]

Firm MFP is constructed using the OECD-ORBIS database developed by Gonnard and Ragoussis (forthcoming) and Gal (2013) based on the Bureau Van Dijk (BvD) ORBIS dataset. It follows the approach of Woodridge (2009).[14] The panel consists of 11 OECD countries and 22 manufacturing sectors over the time period 2000–2009.[15] The selection is based on the number of observations, the stability of the number of observations over time, and the representativeness of the firm size distribution with respect to national business registries (Gal, 2013).

Environmental dependence is measured as pollution intensity, constructed for each industry using the IPPS Pollution Intensity and Abatement Cost World Bank dataset which collects data for the US manufacturing sector in 1987. Appendix A describes the methodology for the construction of the pollution intensity measure and reports related cross country statistics along with two alternative proxies used (see also Robustness checks section) .

Main results

Results: industry analysis

A tightening in environmental policy stringency is found to have a positive short-term effect on industry-level productivity growth, in countries where the industry is close to the global technological frontier. This can be interpreted as support for the strong version of the Porter Hypothesis on the industry level. The effect diminishes as the distance to the frontier increases and becomes insignificant far from the frontier.

Table 1 reports the results across the different specifications, with different fixed effect structure as specified in the table. Frontier observations are excluded from the sample in all the specifications.

In line with previous literature, the coefficient on the MFP growth of the leader is positive, indicating a pass-through effect from the leader to the lagging industries. Moreover, there is evidence of catch-up effects (positive and significant coefficient of the gap), namely country-industry pairs that are further from the technology frontier tend to grow faster. The time trend captures the slowdown of MFP growth in advanced economies (Gordon, 2012), while the business cycle and lagged R&D expenditure are positively correlated with productivity growth. Other controls have expected signs, but turn out mostly insignificant.

Across the specifications the tightening of environmental policy is associated with a short-term, positive effect on industry-level productivity growth and with a negative effect coming from the interaction term with the distance to frontier. Therefore, the overall estimated marginal effect of a tightening in environmental policy on industry productivity growth depends on the technological advancement of the country-industry pair with respect to the global frontier:

$$\frac{\partial \Delta \ln MFP_{cit}}{\frac{1}{3}\sum_{j=1}^{3}\left(ED_{i1987}\Delta EPS_{ct-j}\right)} = \alpha_2 + \alpha_3 gap_{cit-1}$$

To check for the magnitude and significance of the estimated marginal effect we calculate the annual MFP effect for high polluting industries over the observed distance to frontier range, including the corresponding confidence intervals, for an average change of EPS of 0.12 points, which is the mean in-sample change of EPS. Fig. 2 shows that the annual effect of tightening in industry-EPS on industry level MFP growth is positive throughout most of the distance to frontier, leading overall to a permanently higher MFP level. This effect fades away only for countries far from the frontier.

This heterogeneous effect suggests that in each industry the most productive countries benefit most in terms of productivity growth. One reason may be that in these countries firms have access to top technologies and are most capable to adapt to the new regulations, for example by improving production technology. They may also have the best access to financial markets, networks etc., hence being better suited to accommodate the policy change. In this respect, our result points towards the Porter Hypothesis. However, the effect could also be due to a reallocation effect, with productivity gains due to a more rapid axing of less productive firms or outsourcing of less-productive activities as a result of additional costs

[13] The countries included are Australia, Austria, Belgium, Canada, Denmark, Finland, France, Germany, Greece, Italy, Japan, the Netherlands, Norway, Spain, Sweden, the United Kingdom and the United States. The manufacturing industries covered in the sample are food, beverages and tobacco (ISIC Rev.3.1 code 1516), textiles and footwear (1719), wood (2000), pulp, paper and printing (2122), chemical, rubber, plastics and fuel products (2325), other non-metallic mineral products (2600), basic metals and fabricated metal products (2728), machinery and equipment (2933), transport equipment (3435) and manufacturing n.e.c. and recycling (3637).

[14] MFP figures are built based on Levinsohn and Petrin (2003). Woodridge's production function estimation is a one-step procedure that attempts to solve the capital measurement issue (Ackerberg et al., 2006). See Gal (2013) for a detailed and comprehensive discussion of the construction of the MFP measures.

[15] The countries included are: Belgium, Finland, France, Greece, Italy, Japan, Korea, the Netherlands, Spain, Sweden and the United Kingdom. The industry coverage is 2-digit manufacturing (NACE Rev. 1.1 industries 15 to 37). The baseline specification covers 191,597 firms.

Table 1
Industry level: main estimation results.

MFP growth	1 Baseline	2 Baseline	3 Baseline + Regulation	4 Baseline + Regulation	5 Full	6 Full
Leader MFP growth	0.14*** (0.03)	0.16*** (0.03)	0.14*** (0.03)	0.16*** (0.03)	0.12*** (0.03)	0.14*** (0.03)
Gap ($t-1$)	0.089*** (0.01)	0.17*** (0.03)	0.089*** (0.01)	0.17*** (0.03)	0.088*** (0.01)	0.16*** (0.03)
EPS tightening[a]	0.10*** (0.02)	0.11*** (0.03)	0.11*** (0.02)	0.12*** (0.03)	0.12*** (0.02)	0.12*** (0.03)
Gap ($t-1$)*EPS tightening[a]	−0.14** (0.07)	−0.17* (0.09)	−0.14** (0.07)	−0.18* (0.09)	−0.15*** (0.05)	−0.16** (0.07)
Output gap	0.36*** (0.09)	0.37*** (0.10)	0.38*** (0.10)	0.39*** (0.10)	0.40*** (0.09)	0.41*** (0.10)
Year trend	−0.0024*** (0.00)	−0.0025*** (0.00)	−0.0025*** (0.00)	−0.0025*** (0.00)	−0.0029*** (0.00)	−0.0030*** (0.00)
Crisis	−0.0029 (0.01)	−0.0010 (0.01)	−0.0030 (0.01)	−0.0012 (0.01)	−0.0043 (0.01)	−0.0028 (0.01)
EPL ($t-1$)			0.012 (0.01)	0.013 (0.01)	0.012 (0.01)	0.013 (0.01)
PMR ($t-1$)			−0.0093 (0.01)	−0.0073 (0.01)	−0.0067 (0.01)	−0.0037 (0.01)
R&D intensity ($t-1$)					0.0012** (0.00)	0.0035** (0.00)
Constant		4.94*** (1.17)		4.89*** (1.17)		5.96*** (1.20)
Fixed effects						
Country*Industry	No	Yes	No	Yes	No	Yes
Country	Yes	No	Yes	No	Yes	No
Industry	Yes	No	Yes	No	Yes	No
N	2084	2084	2084	2084	1954	1954
Adjusted *R*-squared	0.165	0.115	0.166	0.115	0.184	0.117

Note: Robust standard errors in parentheses and they are clustered at country-industry level.
*** Denotes statistical significance at the 1% level.
** Significance at 5% level.
* Significance at 10% level.
[a] Denotes the moving average of the EPS change over three-year, lagged and multiplied by pollution intensity (ED).

imposed by environmental policies. This reallocation and entry-exit effect is not identifiable at the industry level, but would likely inflate the observed productivity gains. To shed further lights on these dynamics we (i) study the effects of a tightening of environmental policy at the firm level, and (ii) we compare the aggregate-firm and industry results over a common sample (time and countries).

Results: firm analysis

The positive finding at the industry level is only partially confirmed in the firm analysis. Table 2 reports the results for the main specification at the firm level, which augments the industry model with firm level controls (see section Econometric model). Overall, only one-fifth of the firms are estimated to reap productivity gains after a change in environmental

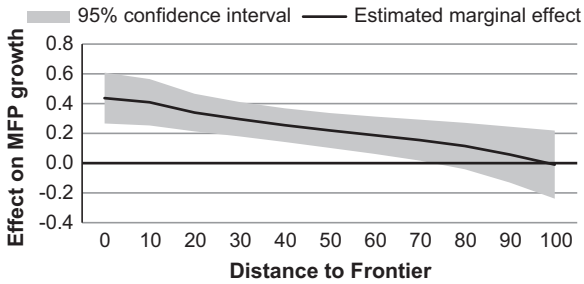

Fig. 2. Effect of an average EPS tightening for high polluting industries by the distance to frontier. *Note:* Annual MFP Effect of a 0.12 point increase in environmental policy stringency (mean in-sample change of EPS) for highly polluting industries. 95% confidence intervals are reported.

regulation, while about a half of the least productive firms in the sample face a negative effect. Fig. 3 reports the annual effect of a mean tightening of EPS on firm multifactor productivity growth, for firms in high polluting industries.

Comparison industry versus firm results

Fig. 4 reports the estimated effects of a change in environmental policy on industry and firm level productivity growth disentangling the effect as function of two dimensions: country/firm productivity (distance to frontier) and industry pollution intensity. In particular, the chart shows the point estimate and the confidence interval for a change in the EPS index equal to the mean cross-country tightening over the period considered. The left hand panel refers to the industry case: highly productive countries experience a temporarily increased MFP growth regardless of the level of pollution intensity. The right hand panel reports firm-level results for the same category matrix (high versus low pollution; high versus low productivity) with less productive firms experiencing a slowdown in MFP, especially if they belong to pollution intensive industries.

The significant negative effect for almost half of the firm sample is not found for country-industry pairs in the industry level analysis. This discrepancy may be the outcome of at least two different mechanisms. First, it may come from the difference in sample composition between industry and firm-level analyses either in terms of countries or years considered. The first two columns in Table B1 (Appendix B, industry analysis) and Table B2 (Appendix B, firm analysis), report the estimations over a common year and country subsample and show that the different samples have little role in explaining the industry-firm differences – the results are similar to those obtained in the original samples.

Second, the discrepancy between industry and firm-level results may come from entry/exit dynamics of firms. Particularly the less technologically advanced firms may end up exiting the market or reducing activity scale. The exit of the least efficient firms would raise overall industry productivity, offsetting the negative productivity effects observed in surviving less-technologically advanced firms. Testing for entry/exit dynamics is problematic due to the lack of data. ORBIS includes information on the incorporation date, but this variable suffers from some measurement issues (Gal, 2013). At the same time, exit is not observable as the company may remain active but no longer be included in the database. Keeping these shortcomings in mind, we find some evidence that firm in-sample survival (proxied by the age of a firm) is negatively and significantly correlated with the distance to the technological frontier – which points to a higher turnover among the firms further away from the frontier and implies that the entry/exit story is plausible.[16]

Potential channels

The heterogeneity found at the firm level, may be the outcome of different channels of transmission of environmental policy. A potentially important channel is via innovation, i.e. the 'weak version' of the Porter Hypothesis. The increase in innovation activity can then potentially lead to efficiency gains and productivity increases (strong PH). Additional tests of this channel did not allow us to draw a decisive conclusion, and were seriously limited by the lack of adequate data. Two recent and focused studies report mixed findings on the weak version of the PH: no effects of environmental policies on patenting (Franco and Marin, 2015) or positive significant effects (Rubashkina et al., 2015).

In any case, these results must be treated with caution. There are various problems of using patents as a proxy for innovation, more generally and in this particular exercise. Most "innovations" are not patented – firms tend to introduce innovations that improve production processes or products, without necessarily patenting them; partly because this is costly, partly because many of the innovations are not really patentable. In practice, this means that most firms in the population do not patent at all (in our sample, just over 1% of firms patent in each year). Moreover, the transmission channel of environmental policy through patented (or any break-through) innovation might be subject to relatively long time lags. In fact, the process from the incentive (policy change) to increased innovation efforts (e.g. R&D spending), actual development

[16] The incorporation data are not available for each firm, which reduces the sample to 944,061 observation and 162,073 firms. In our sample, the unconditional correlation between age and technological advancement is 0.18 (p-value of 0.00). Additionally, looking at the average age of firms across the percentiles of the distribution of the distance to frontier, we find a negative trend.

Table 2
Firm level: main estimation results.

MFP growth	1 Baseline	2 Baseline	3 Baseline + Regulation	4 Baseline + Regulation	5 Full	6 Full
Leader MFP growth	0.14*** (0.02)	0.16*** (0.02)	0.14*** (0.02)	0.17*** (0.02)	0.13*** (0.02)	0.16*** (0.02)
Gap ($t-1$)	0.21*** (0.02)	0.24*** (0.01)	0.20*** (0.02)	0.24*** (0.01)	0.21*** (0.03)	0.28*** (0.01)
EPS tightening[a]	0.42*** (0.09)	0.46*** (0.10)	0.39*** (0.09)	0.43*** (0.10)	0.34*** (0.08)	0.34*** (0.09)
Gap ($t-1$)*EPS tightening[a]	−0.19*** (0.05)	−0.22*** (0.06)	−0.18*** (0.05)	−0.21*** (0.06)	−0.15*** (0.05)	−0.18*** (0.05)
Output gap	1.78*** (0.37)	1.91*** (0.37)	1.50*** (0.32)	1.54*** (0.31)	1.96*** (0.38)	2.03*** (0.36)
Year trend	−0.018*** (0.00)	−0.020*** (0.00)	−0.018*** (0.00)	−0.022*** (0.00)	−0.021*** (0.00)	−0.026*** (0.00)
Crisis	−0.043*** (0.01)	−0.043*** (0.01)	−0.042*** (0.01)	−0.044*** (0.01)	−0.054*** (0.01)	−0.059*** (0.01)
EPL ($t-1$)			0.018 (0.07)	0.0051 (0.08)	0.0016 (0.07)	−0.037 (0.08)
RI ($t-1$)			−0.63 (0.44)	−1.33* (0.71)	−1.25** (0.48)	−2.68*** (0.70)
R&D intensity ($t-1$)			0.0040*** (0.00)	−0.0044** (0.00)	0.0047*** (0.00)	−0.0055*** (0.00)
Size ($t-1$)					0.028*** (0.01)	0.039*** (0.00)
Asset turnover ($t-1$)					0.0085* (0.01)	0.017*** (0.00)
Constant		39.8*** (4.24)		44.3*** (6.85)		57.6*** (7.31)
Fixed effect						
Country	No	Yes	No	Yes	No	Yes
Industry	Yes	No	Yes	No	Yes	No
Country*Industry	Yes	No	Yes	No	Yes	No
N	1,578,262	1,578,262	1,473,422	1,473,422	1,062,460	1,062,460
Adj. *R*-squared	0.104	0.123	0.104	0.123	0.104	0.132

Note: Robust standard errors in parentheses and they are clustered at country-industry level.
*** Denotes statistical significance at the 1% level.
** Significance at 5% level. * Significance at the 10% level. * Significance at the 10% level.
[a] Denotes the moving average of the EPS change over three years, lagged and multiplied by pollution intensity.

of the technology, patenting it and, finally, adoption which has commercial effects (productivity), is likely to take many years (see David and Wright, 2003; Andrews et al., 2014; Popp, 2015). Therefore, the impact that we find on multi-factor productivity is more likely to be the outcome of small process innovations, technological diffusion, resources reallocation and offshoring.

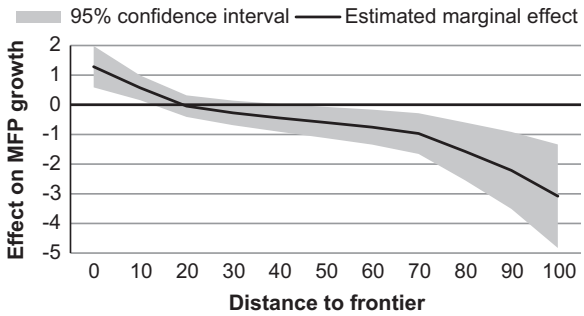

Fig. 3. Effect of an average EPS tightening for firms in high polluting industries by the distance to frontier. *Note:* Annual MFP Effect of a 0.12 point increase in environmental policy stringency (mean in-sample change of EPS) for highly polluting industries. 95% confidence intervals are reported.

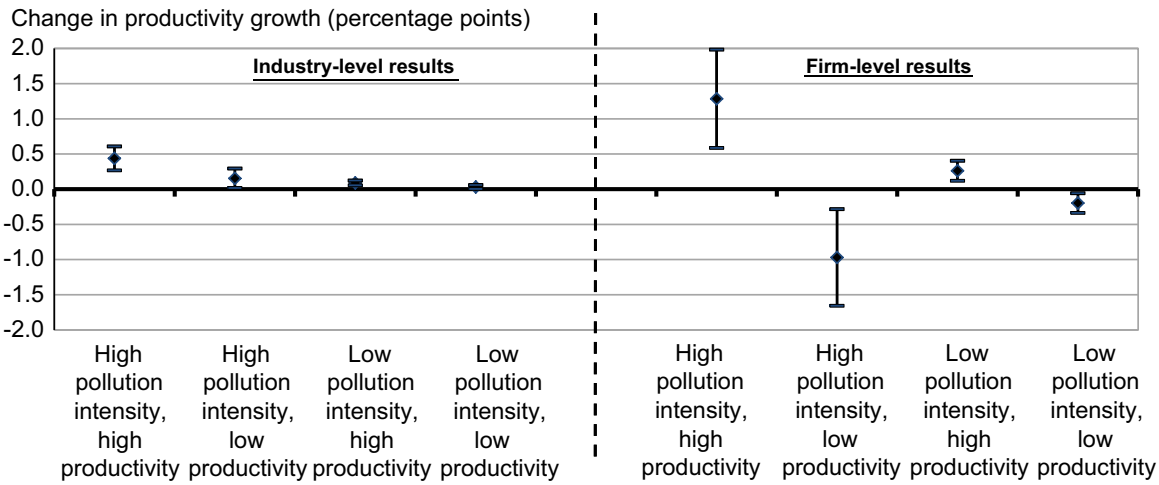

Fig. 4. Effects of a tightening of Environmental Policy Stringency on productivity growth Industry and firm level – economic significance. *Note:* (1) One year effects of a mean in-sample increase in environmental policy stringency, i.e. 0.12 change in the value of the EPS index in one single year. Effects are estimated to last for three years after the policy change and then fade away. No lead effect is found. (2) High (low) pollution intensity is defined as an industry with the highest (lowest) pollution intensity on seven selected key pollutants with respect to value added. (3) High productivity is defined as the country-industry pair (or firm) on or close to the estimated global industry (or firm) productivity frontier. Low productivity is defined as country-industry pair (or firm) at the 70th percentile of distance to the global industry (or firm) productivity frontier. 90% confidence intervals are reported.

The most productive firms are more likely to be involved in international trade or be part of a multinational group. Therefore, they have more resources and capacity to adapt rapidly to changes in the regulation, to innovate, exploit intra-group R&D and know-how in efficient and cleaner technologies, which lowers costs and speeds up technological implementation. Therefore, highly productive firms may be best suited to profit rapidly from changing conditions, seizing new market opportunities, rapidly deploying of new technologies or reaping previously overseen efficiency gains as implied by the Porter Hypothesis. They are also likely to be in a better position to offshore part of the production or substitute inputs profiting from access to global value chains. The role of the internationalization channel is confirmed by Albrizio (forthcoming), who uses a similar specification on the same dataset, and finds a stronger (positive) effect of a tightening of EPS for multinationals than for domestic firms.

Additional hypotheses tested

We use our datasets to test two additional hypotheses: the importance of EPS levels and the differential effect of market versus non-market EPS instruments.

The economic effect of the same change in environmental policy stringency may depend on whether a country already has lax or tight environmental policies. On the one hand, further tightening in an already highly regulated country can be expected to be more detrimental than in a set-up where almost no regulation exists. On the other hand, the marginal effect of the tightening may be smaller if environmental policies are already stringent. To verify these conjectures the change in the EPS variable is interacted with a dummy indicating whether the country is above or below the sample median level of

environmental policy stringency. No evidence is found that the country's level of environmental policy stringency affects the results: the temporary effect in productivity growth resulting from policy tightening is similar across countries, regardless of whether they have a high or low level of environmental regulation.[17]

The *narrow version* of the Porter Hypothesis suggests that it is more likely that environmental policies, which focus on the outcome rather than on the production process, lead to more innovation or higher productivity. This boils down to a differentiation of market-based versus non-market environmental policies. Market-based environmental policy instruments, which are based on explicit price signals, are generally considered to be more cost effective than non-market ones (De Serres et al., 2010). For instance, under an emission trading scheme firms have more flexibility in choosing the technology and timing of adjustment than in a scheme of rigid technology standards. Hence, firms subject to market-based instruments can be expected to experience less detrimental effects on productivity growth. To investigate whether the effect of EPS changes on MFP growth varies with the design features of the environmental policy instruments, the country-level EPS index is split into market-based and non-market policies. Empirical results across the two level of analysis yield some support for the narrow version of the Porter Hypothesis. At the firm level, the market-based instruments show a positive significant coefficient whereas the non-market instruments have no significant effect on productivity growth when using the full set of controls (Appendix B, Table B2, column 2). The effect of the interaction with the distance to frontier is negative and strongly significant for the market component, while less robust for the non-market one. The average annual effects differ considerably for the two groups of instruments (Fig. 5).

At the industry level (Appendix B, Table B1, column 2) the effect of non-market policies is still significant. The magnitude of the estimates is similar across the two sub-components. The interaction term with the distance to frontier is not robust anymore for the market-based instrument, suggesting that market-based instruments do not penalize less productive industries.

Robustness checks

Three further robustness checks were performed at each level of analysis (industry and firms). These specifications test (i) the robustness of our results to the choice of the policy exposure variable, namely the environmental dependence measure, (ii) the ability of the EPS indicator to represent economy-wide environmental policies and (iii) the importance of cross-country differences in policy enforcement. Additional tests supporting the proposed identification are also provided.

Sensitivity to the choice of the exposure variable

Appendix B reports the re-estimated results when alternative definitions of environmental dependence are used: the industries energy dependence (in sample) and energy cost share (pre-sample). In each of the cases the results are not significantly affected, implying the results do not hinge on the exact definition of environmental dependence applied (Appendix B, Table B1, columns 3 and 4; Appendix B, Table B2, columns 3 and 4).[18]

Representativeness of the EPS proxy

The identification strategy is based on the assumption that overall stringency of environmental regulations can be approximated by looking at policy instruments that regulate environmental externalities in selected sectors. A large part of the policy instruments used for constructing the EPS proxy relate primarily to fossil-fuel electricity generation; hence the EPS measure may be less representative for countries with a low share of fossil fuel based electricity generation. To the extent such countries may score lower than expected on the EPS proxy (e.g. lower stringency of the particular policies captured by the EPS proxy), this may imply erroneous estimates. However, re-estimating the equations excluding countries that have a fossil-fuel electricity generation capacity share below 30% (Norway, France, Sweden and Canada) does not affect our results significantly (Appendix B, Table B1, column 5; Appendix B, Table B2, column 5).

Differences in policy enforcement

The EPS indicator is a *de jure* measure and does not capture differences across countries in the implementation and enforcement of policies. The responses to the World Economic Forum's survey on how environmental regulation is enforced can be used to test for the importance of the differences in policy enforcement. We divide countries into high (dummy = 1) and low (dummy = 0) enforcement countries, relative to the median value over the entire available sample. This dummy has no time dimension, hence is interacted with the EPS. At both layers of analysis, the interaction term is not significant,

[17] Complete results can be obtained from the authors upon request.

[18] Note that one of the channels through which regulation may affect industry and firm multifactor-productivity is by rising energy prices. However, our aim is to quantify the overall effect, rather than its directness.

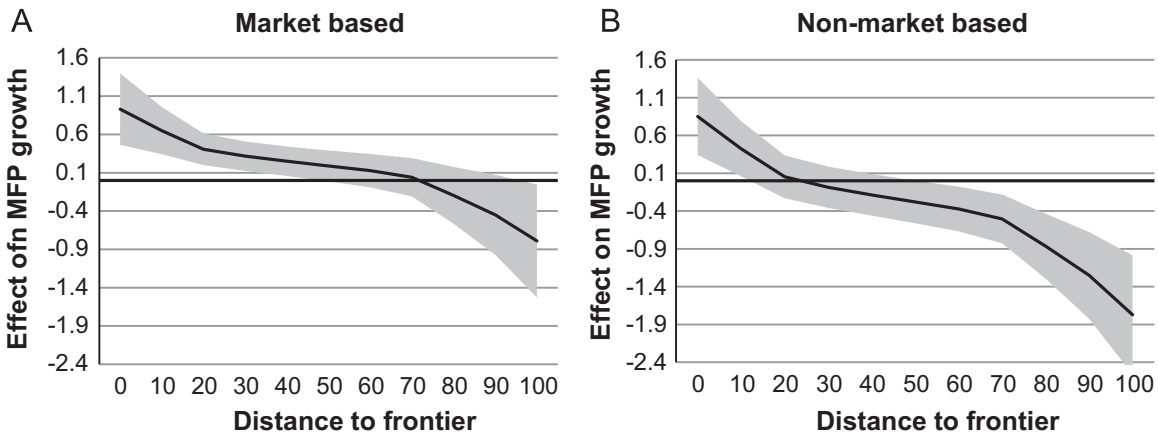

Fig. 5. Average annual effect: market-based versus non-market EPS tightening – firm level. *Note*: Panel A shows the effect of a 0.12 point increase in the stringency of market based instruments (mean in-sample EPS change), for highly polluted industries. Panel B shows the MFP effect of a 0.12 point increase in the stringency of non-market based instruments (mean in-sample EPS change). 95% confidence intervals are reported.

suggesting no additional effects, while the main explanatory variables remain significant and stable in term of the coefficient estimate (Appendix B, Table B1, column 6; Appendix B, Table B2, column 6).

Endogeneity concerns

Potential endogeneity of the EPS index due to reverse causality or simultaneity may question our identification strategy. This could be a problem if, for instance, good performance in given industries (in terms of MFP growth) facilitates adoption of more stringent environmental policies or if firms that are performing poorly are able to successfully lobby against more stringent policies. In order to reduce these concerns we adopt the following precautions. First, the EPS index used is a proxy of overall stringency, based largely on out-of-sample upstream sectors, which makes it less likely that specific industries and firms are able to directly influence the regulatory process. Second, we consider lagged values of the index, which reduces the simultaneity issues. Third, the results are robust to the different fixed effect structures used to control for unobservable confounding factors. Moreover, we explicitly test the potential endogeneity of the EPS index by assessing whether past industry MFP growth can predict current levels of EPS. In a simple regression framework we control for fixed effects (time, industry, country), with a general aim to account for factors such as political economy potentially driving the changes in EPS in response to past industry performance. None of the lags of MFP growth can predict the current changes in EPS. Finally we run an ordered probit estimation on macro-level data to test whether past MFP growth can help to forecast future EPS changes (based on methodology from Mertens and Ravn (2012), among others). The indicator variable takes the value one if there is a tightening of EPS, zero if there is no change and minus one if there is a negative change. Assuming that the regulator would base policy stringency decisions on the level of the emissions, the growth of the economy and on technical change, we include the additional covariates to the model: lagged economic growth (GDP growth rate), lagged greenhouse gas emissions, shifts in the technological frontier capturing the availability of new and more efficient technologies (proxied by lagged MFP growth of the leader – top performer country by year), as well as the domestic ratio of green patent applications (relative to total, 2-year moving average) a variable to account specifically for the innovation effort in cleaner technologies. Country fixed effects are included and one to two-lags specifications are tested. Results lead to rejecting the hypothesis that past MFP growth can help to explain future EPS changes (Table 3). On the contrary, past innovation efforts (green patents), and past economic growth seem good predictors of future environmental policy tightening, as expected.

Conclusion

Over the last twenty years, governments in OECD countries have implemented a wide range of environmental policies with the aim of improving environmental conditions. Inevitably, these policies affect production processes, resource reallocation, capital investment, labor intensity and innovation incentives. Therefore, the ongoing tightening of environmental regulation is likely to affect not only environmental outcomes, but also economic performance. According to the strong version of the Porter Hypothesis environmental policy may lead to higher overall productivity of firms.

We test the effect of a change in environmental policy stringency on productivity growth at two different aggregation levels (industry and firm) in a panel of OECD countries over the last 20 years. We find that a tightening in environmental

Table 3
Ordered probit – macro level.

Indicator variable: EPS tightening											
	GDP growth (L2)	GDP growth (L3)	GHG per capita (L2)	GHG per capita (L3)	MFP growth leader (L2)	MFP growth leader (L3)	MFP growth (L2)	MFP growth (L3)	Green patent ratio	Year	N. Obs
β	0.423*	0.22	0.32	−0.46	−12.25	−9.62	−20.40	−11.49	599.0***	−1.198***	179
SE	0.169	0.163	0.39	0.36	9.62	8.035	12.45	10.69	168.7	0.311	

Note: The dataset considers 23 OECD countries from 1990 to 2009. Data included are: country multi-factor productivity index (Johansson et al., 2013) from which the MFP growth of the leader is calculated as the country with the highest MFP in each year; the ratio of green patents over the total patents application in each country (OECD); per capita GHG emissions is expressed in billions of tons of CO_2 equivalent (OECD). *** Denotes statistical significance at the 1%-level. ** Significant at 5%-level. * Significant at 10%-level.

policy stringency is associated with a subsequent short-run increase in productivity growth for the most productive industries and firms, providing empirical support for the strong version of the Porter Hypothesis. However, this positive effect may come from process improvement as well as from offshoring. At the same time, less productive firms experience a fall in MFP growth, suggesting that the increase in productivity found at the industry level partly reflects the exit of least productive firms from the market. The effect on productivity growth does not depend on whether countries already have a high or low level of environmental regulation.

Finally, market-based environmental policy instruments are found slightly more productivity growth friendly than non-market instruments, as suggested by the narrow version of the Porter Hypothesis. This may be because they provide firms with higher flexibility in the abatement process, by allowing them to choose either the most suitable technology solution or the timing of the adjustment.

Appendix A. Data description

See Table A1.

Table A1
Descriptive statistics.

Industry dataset					Firm dataset				
	MFP growth			EPS[a]		MFP growth			EPS[a]
Industry	**Mean**	**Std. dev.**	**Freq.**	**3-y MA**	**Country**	**Mean**	**Std. dev.**	**Freq.**	**3-y MA**
1516	0.003	0.062	205	0.082	BEL	0.009	0.304	110,397	0.076
1719	0.020	0.055	212	0.066	ESP	−0.019	0.327	556,936	0.112
2000	0.012	0.068	208	0.114	FIN	0.046	0.281	47,561	0.096
2122	0.015	0.045	209	0.043	FRA	0.014	0.256	244,648	0.097
2325	0.022	0.060	205	0.118	GBR	0.025	0.457	72,537	0.075
2600	0.012	0.057	209	0.102	GRC	−0.021	0.378	34,728	0.041
2728	0.012	0.061	209	0.112	ITA	−0.035	0.368	388,681	0.060
2933	0.038	0.069	211	0.041	JPN	0.002	0.101	61,277	0.029
3435	0.017	0.073	207	0.046	NLD	0.023	0.394	8693	0.122
3637	0.011	0.084	209	0.024	SWE	0.044	0.095	141,810	0.127
Total	0.016	0.065	2084	0.075	Total	−0.006	0.317	1,667,268	0.090

Note: [a] denotes the moving average of the weighted EPS*ED change over three years, averaged across the time period considered. The number of observations varies depending on the controls included in the specification. These statistics consider the basic specification, as reported in the industry and firm model section.

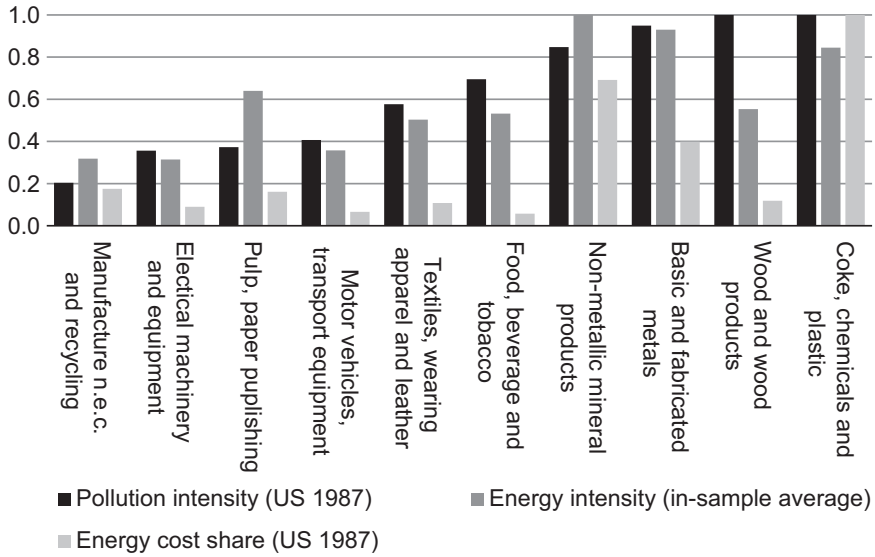

Fig. A1. Exposure variable by industry.
Source: Authors calculation based on WB pollution intensity data on US manufacturing sectors for 1987, the OECD Input-Output (average across vintages), and the ratio between energy inputs and intermediate inputs at current purchasers' prices (both in millions of US Dollar) for US manufacturing sector in 1987. All three indicators are rescaled between 0 and 1.

Environmental dependence

Industries are ranked based on pollution intensity (relative to value added) on seven pollutant categories (two water pollutants, four air pollutants, one toxic substance). The "environmental dependence" is then the simple average of these seven scores, and it can take values from zero (least polluting industry) to 1 (most polluting industry). We use two alternative proxies for industry exposure to environmental policy: the energy dependence index based on OECD STAN Input-Output tables, and the energy cost share based on World KLEMS tables for US 1987 (Fig. A1). The robustness checks are reported in Appendix B, Table B1, columns 3 and 4, and Appendix B, Table B2, columns 3 and 4. Details on the advantages and disadvantages of each of the environmental dependence measures are discussed in Albrizio et al. (2014).

Appendix B. Additional hypothesis tested

See Tables B1 and B2.

Table B1
Industry level: robustness checks and additional hypothesis tested.

MFP growth	1 **Common sample industry and firms**	2 **Market versus non-market instruments**	3 **Exposure variable: energy dependency**	4 **Exposure variable: energy cost share**	5 **Representativeness of the EPS proxy**	6 **Differences in policy enforcement**
Leader MFP growth	0.11**	0.12***	0.13***	0.13***	0.13***	0.12***
	(0.05)	(0.03)	(0.03)	(0.03)	(0.03)	(0.03)
Gap ($t-1$)	0.11***	0.088***	0.090***	0.086***	0.082***	0.088***
	(0.02)	(0.01)	(0.01)	(0.01)	(0.01)	(0.01)
EPS tightening^{o}	0.089***				0.13***	0.096***
	(0.03)				(0.02)	(0.03)
Gap ($t-1$)*EPS tightening^{o}	−0.16**				−0.16***	−0.14**
	(0.07)				(0.05)	(0.06)
Output gap	0.54***	0.39***	0.40***	0.38***	0.39***	0.42***
	(0.19)	(0.09)	(0.09)	(0.09)	(0.11)	(0.10)
Year trend	−0.0028**	−0.0028***	−0.0029***	−0.0025***	−0.0029***	−0.0029***
	(0.00)	(0.00)	(0.00)	(0.00)	(0.00)	(0.00)
Crisis	−0.012	−0.0038	−0.0045	−0.0075	−0.0011	−0.0042
	(0.02)	(0.01)	(0.01)	(0.01)	(0.01)	(0.01)
EPL ($t-1$)	0.013	0.011	0.013	0.014	0.018	0.013
	(0.06)	(0.01)	(0.01)	(0.01)	(0.01)	(0.01)
PMR ($t-1$)	−0.0012	−0.0070	−0.0072	−0.0044	−0.0076	−0.0064
	(0.01)	(0.01)	(0.01)	(0.01)	(0.01)	(0.01)
R&D intensity ($t-1$)	0.00091	0.0012**	0.0012**	0.0011**	0.0013*	0.0012**
	(0.00)	(0.00)	(0.00)	(0.00)	(0.00)	(0.00)
Market EPS tightening^{o}		0.096***				
		(0.02)				
Gap ($t-1$)*M EPS tightening^{o}		−0.12***				
		(0.04)				
Non-Market EPS tightening^{o}		0.027				
		(0.02)				
Gap ($t-1$)*NM EPS tightening^{o}		−0.038				
		(0.04)				
EPS tightening^{o} (En. Dep)			0.14***			
			(0.03)			
Gap ($t-1$)*EPS tightening^{o} (En.Dep)			−0.16***			
			(0.06)			
EPS tightening^{o} (En.Cost)				0.42***		
				(0.11)		
Gap ($t-1$)*EPS tightening^{o} (En.Cost)				−0.46**		
				(0.19)		
EPS tightening^{o}*Enforcement						0.028
						(0.04)
Gap ($t-1$)*EPS tightening^{o}*Enforcement						−0.0055
						(0.08)
Fixed effects						
Country*Industry	Yes	Yes	Yes	Yes	Yes	Yes
N	833	1954	1954	1954	1399	1954
Adj. *R*-squared	0.170	0.188	0.184	0.179	0.165	0.184

Note: Robust standard errors in parentheses, clustered at country-industry level; ***, ** and * denote statistical significance at the 1%, 5% and 10% levels, respecitvely. ^{o} denotes the moving average of the EPS change over three years ($t-1$ to $t-3$).
Each of the columns represents specifications derived from column (6) in Table 1. Column 1 shows results for a common firm and industry level subsample (overlapping). Column (2) shows results from market-based and non-market instruments; (3) and (4) show results with alternative definitions of the exposure variable; (5) shows results with countries with small shares of fossil-fuel based electricity excluded; and (6) shows results when attempting to account for enforcement.

Table B2
Firm level: robustness checks and additional hypothesis tested.

MFP growth	1 **Common sample industry and firms**	2 **Market versus non-market instruments**	3 **Exposure variable: Energy dependency**	4 **Exposure variable: Energy cost share**	5 **Representativeness of the EPS proxy**	6 **Differences in policy enforcement**
Leader MFP growth	0.13***	0.071***	0.12***	0.12***	0.12***	0.13***
	(0.02)	(0.03)	(0.02)	(0.02)	(0.03)	(0.02)
Gap ($t-1$)	0.21***	0.21***	0.22***	0.21***	0.22***	0.22***
	(0.03)	(0.03)	(0.03)	(0.03)	(0.03)	(0.03)
EPS tightening$^{\varrho}$	0.34***				0.50***	0.42***
	(0.08)				(0.10)	(0.09)
Gap ($t-1$)*EPS tightening$^{\varrho}$	−0.15***				−0.22***	−0.18***
	(0.05)				(0.07)	(0.05)
Output gap	1.96***	4.98***	2.03***	1.81***	1.90***	1.89***
	(0.38)	(0.83)	(0.37)	(0.38)	(0.40)	(0.39)
Year trend	−0.021***	−0.027***	−0.021***	−0.021***	−0.023***	−0.021***
	(0.00)	(0.00)	(0.00)	(0.00)	(0.00)	(0.00)
Crisis	−0.054***	−0.10***	−0.057***	−0.057***	−0.054***	−0.053***
	(0.01)	(0.02)	(0.01)	(0.01)	(0.01)	(0.01)
EPL ($t-1$)	0.0016	−0.23**	−0.017	−0.0056	−0.25***	0.047
	(0.07)	(0.10)	(0.08)	(0.09)	(0.09)	(0.07)
RI ($t-1$)	−1.25**	−0.91*	−1.36***	−1.51***	−1.21**	−1.16**
	(0.48)	(0.46)	(0.48)	(0.48)	(0.53)	(0.47)
R&D intensity ($t-1$)	0.0047***	0.0064***	0.0052***	0.0047***	0.0072*	0.0047***
	(0.00)	(0.00)	(0.00)	(0.00)	(0.00)	(0.00)
Size ($t-1$)	0.028***	0.024***	0.028***	0.028***	0.031***	0.028***
	(0.01)	(0.01)	(0.01)	(0.01)	(0.01)	(0.01)
Asset turnover ($t-1$)	0.0085*	0.015**	0.0096*	0.0091*	0.014***	0.0083
	(0.01)	(0.01)	(0.01)	(0.01)	(0.00)	(0.01)
Market EPS tightening$^{\varrho}$		0.28***				
		(0.07)				
Gap ($t-1$)*M EPS tightening$^{\varrho}$		−0.13***				
		(0.03)				
Non-Market EPS tightening$^{\varrho}$		0.29***				
		(0.05)				
Gap ($t-1$)*NM EPS tightening$^{\varrho}$		−0.080***				
		(0.02)				
EPS tightening$^{\varrho}$ (En. Dep)			0.42***			
			(0.06)			
Gap ($t-1$)*EPS tightening$^{\varrho}$ (En.Dep)			−0.12***			
			(0.02)			
EPS tightening$^{\varrho}$ (En.Cost)				0.52***		
				(0.17)		
Gap ($t-1$)*EPS tightening$^{\varrho}$ (En.Cost)				−0.16***		
				(0.06)		
EPS tightening$^{\varrho}$*Enforcement						−0.16
						(0.11)
Gap ($t-1$)*EPS tightening$^{\varrho}$*Enforcement						0.0034
						(0.05)
Fixed effects						
Country*Industry	Yes	Yes	Yes	Yes	Yes	Yes
N	1,062,460	562,814	1,072,550	1,072,550	850,509	1,062,460
Adj. *R*-squared	0.104	0.105	0.106	0.104	0.105	0.105

Note: Robust standard errors in parentheses, clustered at country-industry level; ***, ** and * denote statistical significance at the 1%, 5% and 10% levels, respecitvely. $^{\varrho}$ denotes the moving average of the EPS change over three years ($t-1$ to $t-3$).
Each of the columns represents specifications derived from column (6) in Table 1. Column 1 shows results for a common firm and industry level subsample (overlapping). Column (2) shows results from market-based and non-market instruments; (3) and (4) show results with alternative definitions of the exposure variable; (5) shows results with countries with small shares of fossil-fuel based electricity excluded; and (6) shows results when attempting to account for enforcement.

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